MACHINE LEARNING

Twenty Seventh Lecture by Prof. Uday S. Dixit, Department of Mechanical Engineering, IIT Guwahati



Machine Learning

- **Machine learning is a branch of artificial intelligence (AI) that enables systems to learn from data and improve their performance without being explicitly programmed.**
- **It involves using algorithms to analyze data, identify patterns, and make predictions or decisions.**
- **Essentially, machine learning systems "learn" from experience as they process more data, becoming more accurate over time**

What is Machine Learning (ML)?

- "The field of study that gives computers the ability to learn on their own" (Samuel, 1959).
- ML algorithms learn the pattern from past datasets and apply the same to new or unseen datasets.
- ML uses probability theory, statistics, linear algebra and calculus for drawing inferences.
- A branch of Artificial Intelligence (AI)
- Allows computers to learn from data without being explicitly programmed
- Uses algorithms to identify patterns and make decisions



Arthur Samuel (1901-1990)

https://commons.wikimedia.org/wiki/File:This_is_the_photo_of_Arthur_Samuel.jpg

Why is ML Important Today?

- Powers many modern technologies: search engines, recommendations, voice assistants
- Automates and improves decision-making
- Helps discover insights from large datasets
- Automates tasks
- Finds patterns in data

Machine Learning (ML) in Manufacturing

Benefits of applying ML in Manufacturing:

- ML algorithms has the ability to handle high dimensional data.
- ML can help discover unknown patterns in dataset that was formerly unaware.
- Manufacturing systems are dynamic and uncertain, ML helps learn from such dynamic environment and help in decision making.

Challenges of applying ML in Manufacturing:

- Acquisition of relevant data
- Preprocessing of the captured data
- A suitable ML technique to apply on the captured dataset
- Interpretation of the results obtained

Need for ML in the current manufacturing scenario

- The following are the major goals of the manufacturer:
  - Higher machining efficiency
  - Lower cost
  - Higher quality of the product
- The main aim of implementing ML in modern day industries is to ensure a cost efficient running without compromising safety.

Need for ML in the current manufacturing scenario (cont.)

- The key performance indicators in machining such as cutting force, cutting tool temperature, surface quality can be directly analysed using past data as well as real time data based on the controllable and uncontrollable variables.

- Proper implementation of big data in manufacturing can drastically reduce the product development time and other cost associated with it as reported in a study conducted by McKinsey Global Institute Research Organization (Manyika, 2011) .

Manyika, J., Chui, M., Brown, B., Bughin, J., Dobbs, R., Roxburgh, C., & Hung Byers, A. (2011). Big data: The next frontier for innovation, competition, and productivity.

How to develop and maintain a databank for applying ML for modern day industries?

- Two examples are presented to propose a methodology on utilising a large amount of past machining data for the enhancement of the productivity.
- The first case involves collection of veracious data from various sources.
- The second case involves efficient storage and prediction.



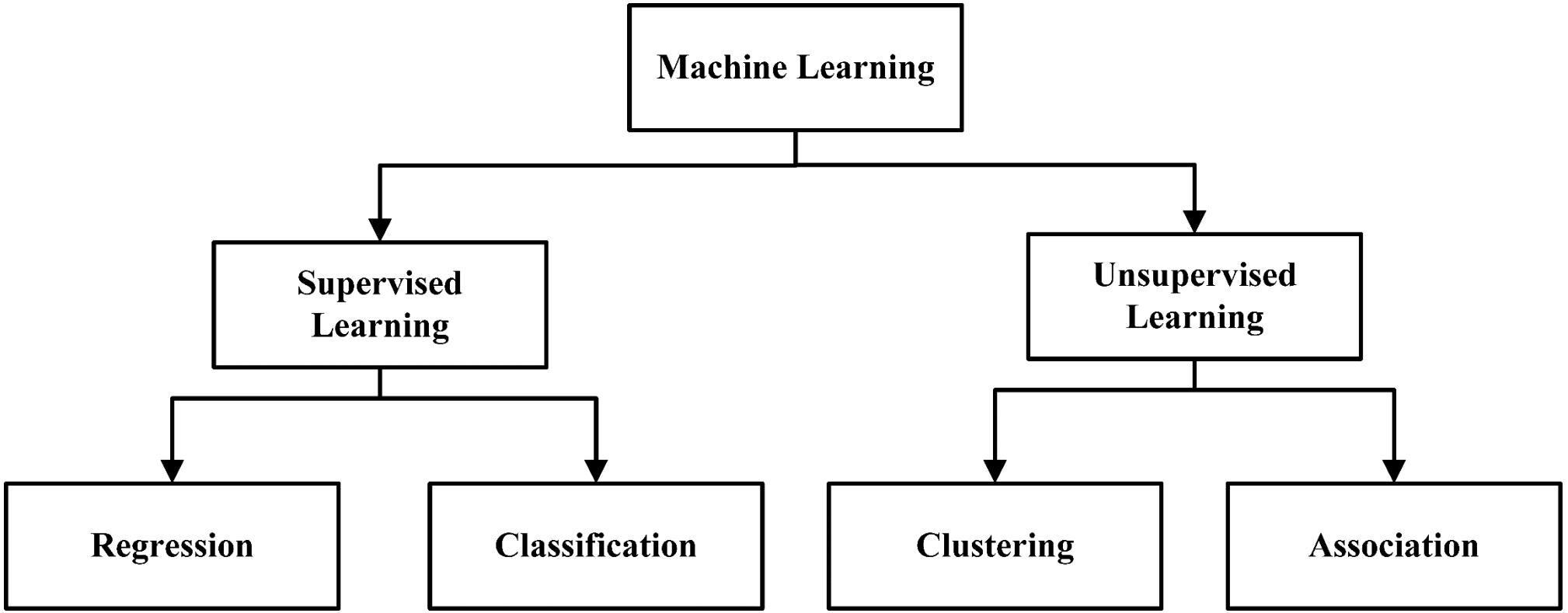
Cloud Server

Types of Machine Learning

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

Reinforcement learning is a type of machine learning where an agent learns to make decisions in a dynamic environment to maximize a reward signal. It's like teaching an AI to play a game or drive a car by rewarding good actions and penalizing bad ones. The agent learns through trial and error, interacting with its environment and receiving feedback based on its actions.

Major types of ML techniques



How Machines Learn?

- Input data → Algorithm → Learn → Predict

ML workflow

- Data Collection
- Data Cleaning & Preprocessing
- Model Training
- Model Evaluation
- Deployment

Data: The Foundation of ML

- **Structured vs Unstructured data**
- **Quality data is crucial**

Supervised Learning

- Learns from labeled data
- Predicts outcomes based on input-output pairs
- Examples: spam detection, price prediction

Unsupervised Learning

- Finds patterns in unlabeled data
- Finds hidden patterns or groupings
- Examples: customer segmentation, Anomaly detection

Reinforcement Learning

- Learns by trial and error
- learns by interacting with environment
- Examples: game playing, robotics, self-driving cars

Example: Teaching a robot to clean a room

Imagine a robot vacuum cleaner that needs to learn how to clean a room efficiently. It doesn't know how to do it at first. So, we use **reinforcement learning** to train it.

1. Environment:

The room, with furniture and dirt in various places.

2. Agent:

The robot vacuum cleaner.

3. States:

Different situations, like: “Robot is near the wall”, “Robot sees dirt under a table”, “Robot is stuck in a corner”

4. Actions:

The robot can: Move forward, Turn left or right, Stop, Start cleaning

5. Reward system:

We define rewards like this:

- +10 points for cleaning a dirty spot
- –5 points for bumping into furniture
- –1 point for wasting battery without cleaning

6. Learning process:

At first, the robot moves randomly. But whenever it **cleans** a spot, it gets **positive reward**.

Whenever it **bumps** into something, it gets a **negative reward**.

Over time, it **learns which actions in which situations give more reward**, and it starts:

- Avoiding obstacles
- Going straight to dirty areas
- Using less battery

Comparison of ML Types

- Supervised: Labeled data → Predict outcome
- Unsupervised: Unlabeled data → Discover structure
- Reinforcement: Learn by trial and error

“Labeled data refers to information that has been assigned one or more labels to provide context and meaning. These labels act as a target for machine learning models, enabling them to learn and make predictions based on the input data. For example, in image recognition, a photo might be labeled as "dog" or "cat," while in spam detection, an email might be labeled as "spam" or "not spam“.

Typical ML Workflow

- Define problem → Collect data → Train model → Test → Deploy

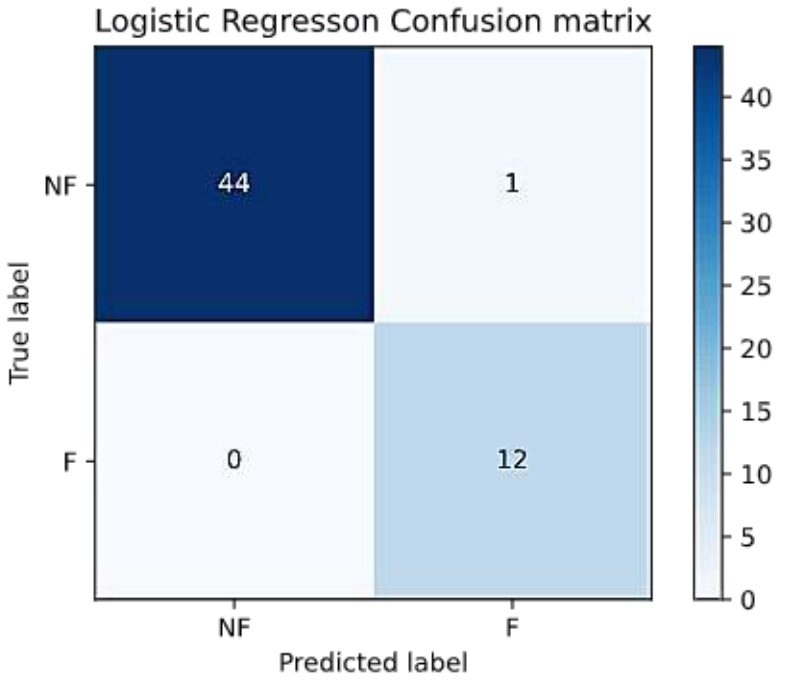
Training vs Testing

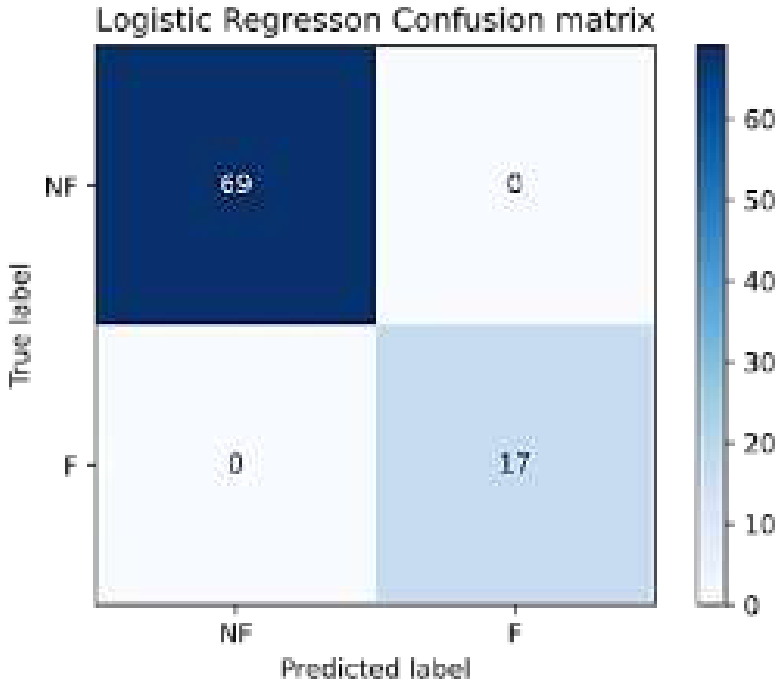
- Training = learn patterns
- Testing = check accuracy

AVOID OVERFITTING.

Model Evaluation Metrics

- Accuracy, Precision, Recall, F1-score
- Confusion Matrix
- Cross-validation





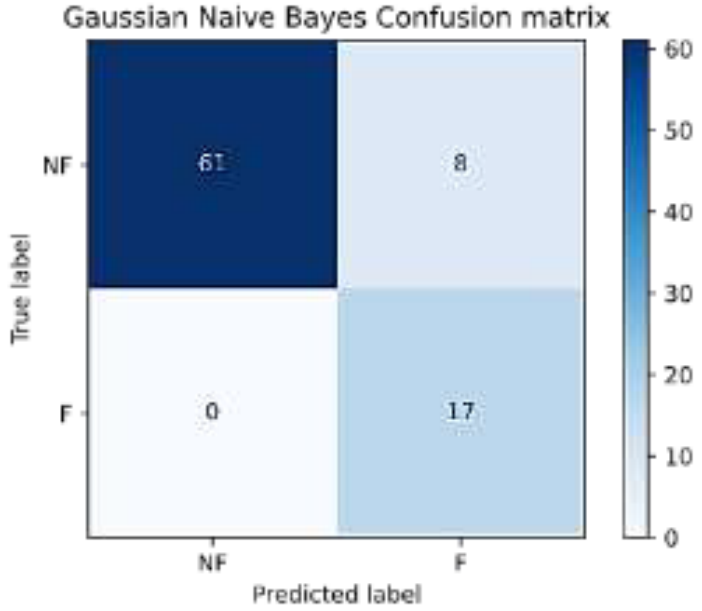


Table 3.1. Performance evaluation metrics

Sr. No.	*Metric*	*Definition*	*Expression*
1	Accuracy	It is a percentage of all cases that have predicted correctly. It is a good measure of overall model performance.	$\frac{TP + TN}{TP + TN + FP + FN}$
2	Precision	It is a percentage of total failure cases predicted that are predicted correctly. A low precision would mean that doing unwanted maintenance frequently.	$\frac{TP}{TP + FP}$
3	Recall (Sensitivity)	It is a ratio of correctly classified failure cases to actual no. of total failure cases. It is used when the aim is to capture the maximum no. of failure cases.	$\frac{TP}{TP + FN}$
4	F-Score	It is a precision and recall harmonic mean. Both precision and recall are summed up in one phrase. As a result, the better the model, the higher the F1 score.	$\frac{2 * (Precision * Recall)}{Precision + Recall}$

Y.V. Jedhe, Planning of machine maintenance based on failure probability estimation, M.Tech. Thesis, IIT Guwahati, 2021.

Supervised Algorithms

- Linear Regression
- Logistic Regression
- Decision Trees
- Random Forest
- k-Nearest Neighbors (k-NN)
- Support Vector Machines (SVM)

Linear Regression

- Predicts continuous values
- Example: Predicting house prices based on size

Logistic Regression

- Used for binary classification (yes/no)
- Example: spam detection (Spam or not spam)

Decision Trees

- Flowchart-like model
- Tree-like model of decisions
- Easy to interpret and visualize

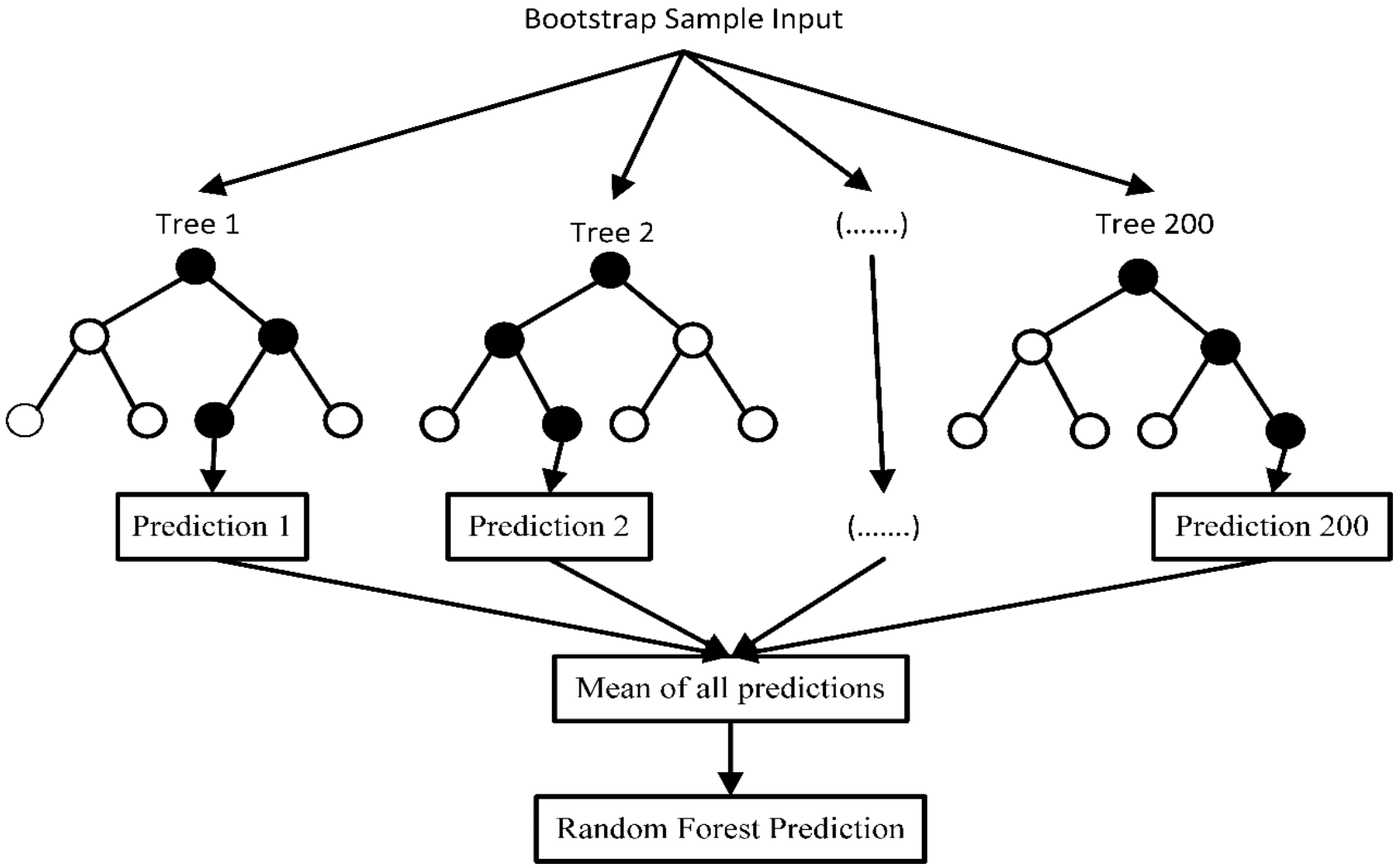
Random Forests

- Ensemble of decision trees
- Reduces overfitting, improves accuracy
- More accurate

Random Forest

- Random Forest falls into the category of ensemble learning.
- Ensemble learning is a technique that combines the predictions made by numerous machine learning algorithms to make it more accurate when compared to individual models.
- An Ensemble model therefore consists of numerous machine learning models.
- Training samples are created from the original dataset termed as Bootstrap sample.
- Bootstrap samples indicates sampling with replacement such that each data has an equal probability of it being selected, and each sample is the same size as the original training dataset.
- Ensemble prediction is defined as the one with the average of all the prediction being made.

Random Forest



The basic structure of Random Forest

K-Nearest Neighbors (KNN)

- Classifies based on closest training examples
- Simple and effective for small datasets
- Looks at closest data points
- Example: image recognition

Support Vector Machines (SVM)

- Finds the optimal boundary between classes
- Effective in high-dimensional spaces
- Finds best decision boundary
- Good for complex problems

Unsupervised Algorithms

- K-means Clustering
- Hierarchical Clustering

K-Means Clustering

- Groups similar data
- Example: customer types

Principal Component Analysis (PCA)

- Reduces data size
- Keeps important information

Principal Component Analysis (PCA) is a dimensionality reduction technique used to simplify datasets by transforming a large set of correlated variables into a smaller set of uncorrelated variables called principal components. These principal components capture the most important patterns and variations in the data.

Reinforcement Algorithms

- Q-learning
- Deep Q-networks (DQN)

Q-learning is a model-free reinforcement learning algorithm that enables an agent to learn optimal actions in an environment through trial and error, maximizing long-term rewards. It achieves this by assigning a Q-value (action-value) to each possible action in a given state, representing the expected reward of that action. The algorithm then iteratively updates these Q-values based on the feedback (rewards) it receives from the environment, gradually converging towards an optimal policy

Applications of ML

Daily Life	Healthcare	Finance	Agriculture	Manufacturing & Engineering
• Voice assistants • Recommendation systems (YouTube, Amazon)	• Disease detection and diagnosis • Medical image analysis • Drug discovery	• Fraud detection • Credit scoring • Customer segmentation	• Yield prediction in agriculture • Smart irrigation systems	• Quality control • Performance estimation in manufacturing • Tool/machine condition monitoring • Fault diagnosis • Image recognition for quality control • Time series forecasting

Challenges in ML

- Data privacy
- Bias in data
- Interpretability of models

Ethics in Machine Learning

- Bias and fairness
- Privacy and data security
- Transparency and accountability